

Prior Design Attempt #1 - Trolley



This trolley for tools sports some designs and features we want to add into our project. The 7 drawers extend out 15 inches, there are wheels for easy mobility, the toolwall contracts back into the trolley like a suitcase. And the sides open for a cabinet-like tool wall. ([Garant Gridline Toolcar, Hoffman Group](#))

Analysis

While this trolley is certainly impressive and provides usability, it comes up short in a few categories. Number one is space. With the abundance of materials and tools Mr Paz has, just the hot glue guns and box cutters alone would take up at least 4 of the 7 drawers. Furthermore, there aren't any safety features that allow students to not injure themselves while trying to grab the box cutters. In all honesty, this design wouldn't be much better than the assortment of boxes Mr Paz uses right now, and could even prove to be less efficient. Future designs could benefit from using

cabinet-like systems instead of drawers, and angled storage compartments on the side for sharper tools.

Prior Design Attempt #2 - Tool chest



This Tool chest has some very useful features to it, all of which can be incorporated into our build. The chest has drawers that vary in size, for a variety of tools. This may sport the most storage out every Prior Design Attempt on this list. Doors for storage on both sides of the chest, along with a big overhead chest, with a total of 15 different drawers, and huge wheels to move it around.

Analysis:

While this design is certainly brilliant, it does fall flat on a few things. The chest is too big and bulky, even with wheels supporting it, the amount of drawers and compartments it has can be described best as a “glutton of materials”. This exceeds budgets by a HUGE margin. All of this is made out of aluminum or steel as well. Even without tools inside it, its mobility is severely limited.

Prior Design Attempt #3: Mobile shelf



This design is made for larger home tools and amenities. It could be used for outdoor purposes like holding plants and tools. We would like to incorporate how sturdy it is as well as how much it can hold.

Analysis:

The shelf is definitely excellent when it comes to space and how much it can carry. It's made of metal so it can carry a lot of weight and is incredibly sturdy. We could possibly incorporate the structural support throughout the build. It does however fall short when it comes to order and organization. The lack of separated compartments results in whatever you place on it just sitting there mixing with everything else and if you do want to organise your materials you would need to buy

eternal containers. This defeats the whole purpose of our projects, to save Dr. Paz time with organisation and set up.

Prior Design Attempt #4 Tool Chest [CRAFTSMAN 1000 Series 26.07-in W x 17.49-in H x 12.1-in D 5 -Drawer Steel Tool Chest \(Black \)](#)



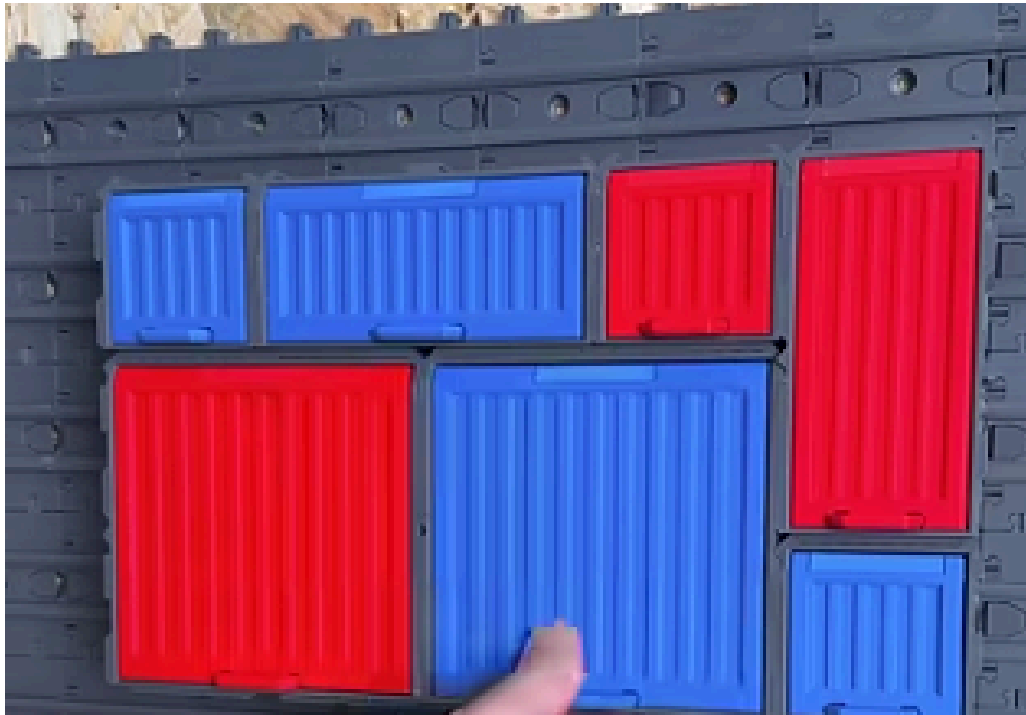
This tool chest has a good size to be able to keep it indoors, it also has varying sizes of drawers and it also has handles on the sides to move it around since it doesn't have wheels. The top is able to pop up like a car hood and has storage for more tools that wouldn't fit in the other compartments.

Analysis:

It is a good way to start visualizing the finished product for Dr. Paz, however, this current model doesn't have wheels, which is a crucial element when creating Dr. Paz's tool organizer since he needs to be able to move it in and out of the storage room.

The top that pops up can be a good idea to incorporate into our design, but we want to make use of that lid so that we can put other tools there.

Prior Design Attempt #5: Infinite Versatile Storage



This infinite versatile storage system is designed with adjustable height and length, with varying size and purpose modules that can be attached to store varied items. The boxes can pull out and tilt for easy access, and the entire system is highly customizable in terms of positioning. It also has the ability to have labeled storage bins.

Analysis:

This system excels in storing multitudes of varied bits and bobs, with a unique customizability that is beyond any ordinary system. However, this system falls short for Dr. Paz since much of what he wants to store is larger than many of these things. Moreover, being made of PLA, the strength of the containers to be scaled is questionable.

